

Use operations confidently in algebraic and financial contexts

Solve each problem carefully. Show your thinking and explain your answer.

Name: _____

Date: _____

Step 11

Upper Elementary

★ Today I am learning to use operations confidently in algebraic and financial contexts.

1. Algebraic Expressions

Write an expression for each situation. Then find the value when the given value is used.

- a) Five more than three times a number n .

Expression: _____

When $n = 8$,

Value: _____



- b) The difference between twice a number x and 7.

Expression: _____

When $x = 12$,

Value: _____



- c) Three times a number y , increased by 15.

Expression: _____

When $y = 6$,

Value: _____



- d) Half of a number m , decreased by 9.

Expression: _____

When $m = 20$,

Value: _____

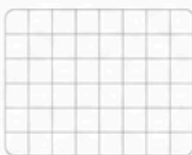


2. Equations and Unknowns

Solve for the unknown. Show your work.

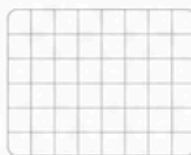
a) $3x + 7 = 25$

$x =$ _____



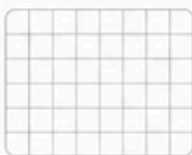
b) $2(y - 4) = 18$

$y =$ _____



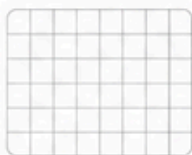
c) $5z - 16 = 34$

$z =$ _____



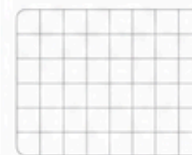
d) $\frac{n}{3} + 9 = 21$

$n =$ _____



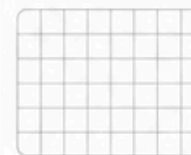
e) $4(p + 5) - 7 = 29$

$p =$ _____



f) $\frac{m}{2} - 6 = 14$

$m =$ _____



3. Patterns and Relationships

Find the rule. Then use it to complete the table and answer the questions.

- a) The table shows a pattern.

Input (n)	1	2	3	4	5
Output	5	8	11	14	17

- i) Write an expression for the rule. _____
 ii) Find the output when $n = 20$. _____
 iii) Find the input when the output is 50. _____

- b) The table shows another pattern.

Input (n)	1	2	3	4	5
Output	2	6	18	54	162

- i) Write an expression for the rule. _____
 ii) Find the output when $n = 6$. _____
 iii) Find the input when the output is 486. _____

4. Financial Contexts

Solve each problem. Show your work.

- a) A notebook costs \$2.75.
A pen costs \$1.45.
Emma buys 3 notebooks and 4 pens.
How much does she spend?



Equation: _____

Answer: \$ _____

- b) A shirt costs \$18.50.
It is on sale for 20% off.
What is the discount?
What is the sale price?



Discount: \$ _____

Sale price: \$ _____

- c) A family has a budget of \$320 for a weekend trip. They spend \$86 on fuel, \$124 on a hotel, and \$57 on food.
How much money do they have left?



Equation: _____

Money left: \$ _____

- d) A tablet costs \$249. It is on sale for \$199, and an extra 10% is taken off at checkout.
What is the final price?



Final price: \$ _____

5. Richer Reasoning

Read carefully. Solve using operations and explain your thinking.

- a) The perimeter of a rectangular garden is 72 meters. The length is twice the width. What are the length and width?



Show your work:

Length: _____ m Width: _____ m

- b) A class has some boys and girls. There are 8 more girls than boys. If there are 28 students in all, how many boys and how many girls are there?



Show your work:

Boys: _____ Girls: _____

- c) Maya has some money. She spends \$14.25 on a book and \$8.60 on a drink. She now has \$27.15 left. How much money did Maya have at first?



Show your work:

Maya had \$ _____ at first.

My Learning Check Circle how confident you feel.



I can do it!



I need more practice.



I need help.

I used...



mental strategies



written methods



properties and rules



calculators (where helpful)



checking my answer

Today I learned...

1 thing I did well: _____

1 thing I will improve: _____